Stage 3: MCP Server Integration for Reservation Processing

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Project

Intelligent Parking Reservation System

Technologies Used

- Python
- FastAPI
- MCP-style Architecture
- LangChain
- JSON Storage
- Requests Library

Objective

Implement a server that processes approved reservations and stores them securely after administrator approval.

Stage 3 Objectives

Goals of Stage 3

- Develop an MCP-style server using FastAPI.
- Process approved reservations automatically.
- Store confirmed reservations in persistent storage.
- Connect Approval Agent with MCP Server.
- Ensure secure and reliable reservation processing.

Expected Outcome

- Automated reservation recording system.
- Seamless communication between agents and server.
- Secure reservation storage mechanism.

MCP Server Workflow

MCP Server Workflow

1. User submits reservation through chatbot.
2. Reservation enters Pending Queue.
3. Administrator reviews request.
4. Reservation approved.
5. Approval Agent sends reservation data to MCP Server.
6. MCP Server writes reservation to storage.
7. Final Result : Confirmed reservation is permanently stored.

Confirmed reservation is permanently stored.

Approval Agent

- Reviews reservations
- Approves or rejects requests

Reservation Writer

- Writes reservation records to text storage

MCP Server (FastAPI)

- Receives approved reservations
- Validates incoming requests
- Processes reservation data

Security & Reliability Features

- Request validation
- Duplicate reservation prevention
- Health monitoring endpoint
- Error handling
- Persistent storage

Storage Layer

- confirmed_reservations.txt

Testing, Results & Future Scope

Testing Performed

Functional Tests

- Reservation writing
- Approval workflow
- MCP API communication

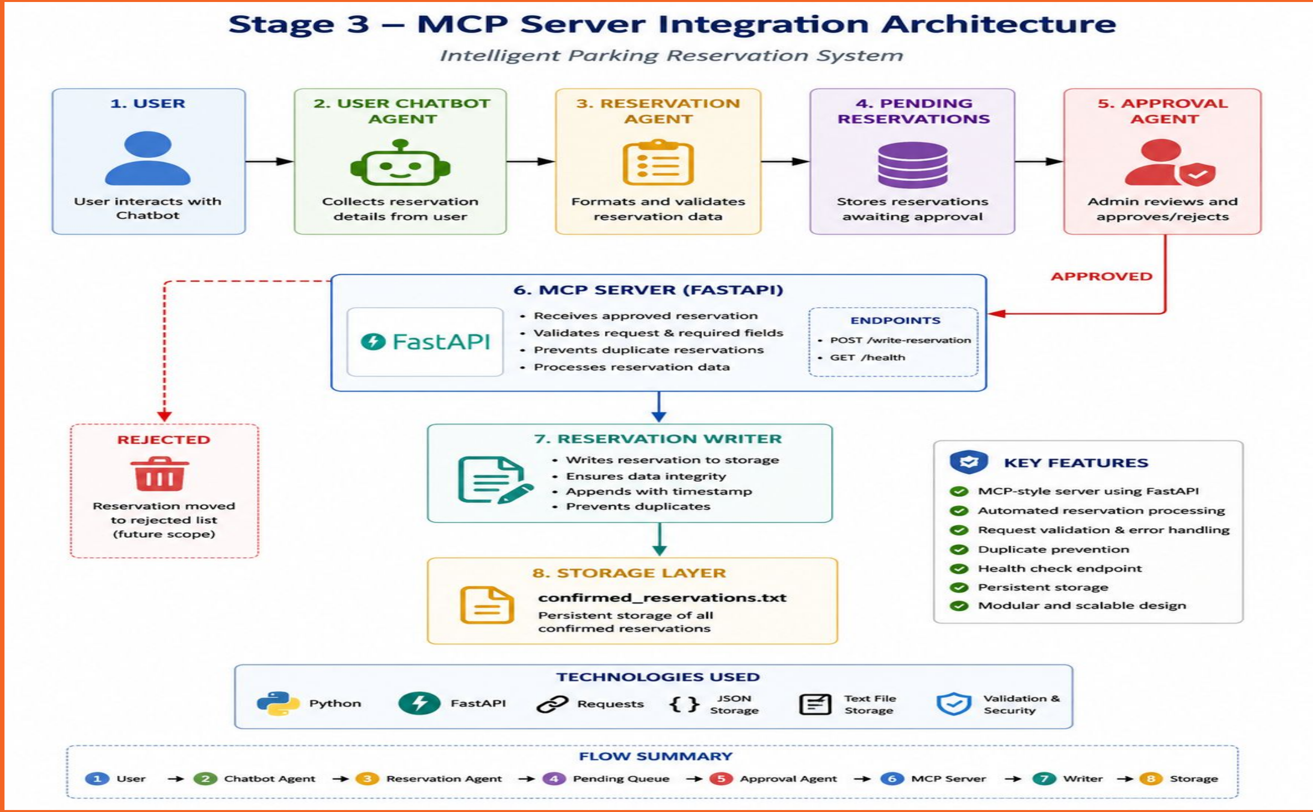
API Tests

- /health
- /write-reservation
- Missing field validation

Results

- Successful end-to-end processing
- Approved reservations stored automatically
- Reliable MCP communication established

ARCHITECTURAL DIG



Stage 3 – MCP Server Integration Architecture
Intelligent Parking Reservation System
1. USER
User interacts with Chatbot
2. USER CHATBOT AGENT
Collects reservation details from user
3. RESERVATION AGENT
Formats and validates reservation data
4. PENDING RESERVATIONS
Stores reservations awaiting approval
5. APPROVAL AGENT
Admin reviews and approves/rejects
APPROVED
6. MCP SERVER (FASTAPI)
FastAPI
• Receives approved reservation
• Validates request & required fields
• Prevents duplicate reservations
• Processes reservation data
ENDPOINTS
• POST /write-reservation
• GET /health
REJECTED
Reservation moved to rejected list (future scope)
7. RESERVATION WRITER
• Writes reservation to storage
• Ensures data integrity
• Appends with timestamp
• Prevents duplicates
8. STORAGE LAYER
confirmed_reservations.txt
Persistent storage of all confirmed reservations
KEY FEATURES
MCP-style server using FastAPI
Automated reservation processing
Request validation & error handling
Duplicate prevention
Health check endpoint
Persistent storage
Modular and scalable design
TECHNOLOGIES USED
Python
FastAPI
Requests
{ } JSON Storage
Text File Storage
Validation & Security
FLOW SUMMARY
1 User → 2 Chatbot Agent → 3 Reservation Agent → 4 Pending Queue → 5 Approval Agent → 6 MCP Server → 7 Writer → 8 Storage

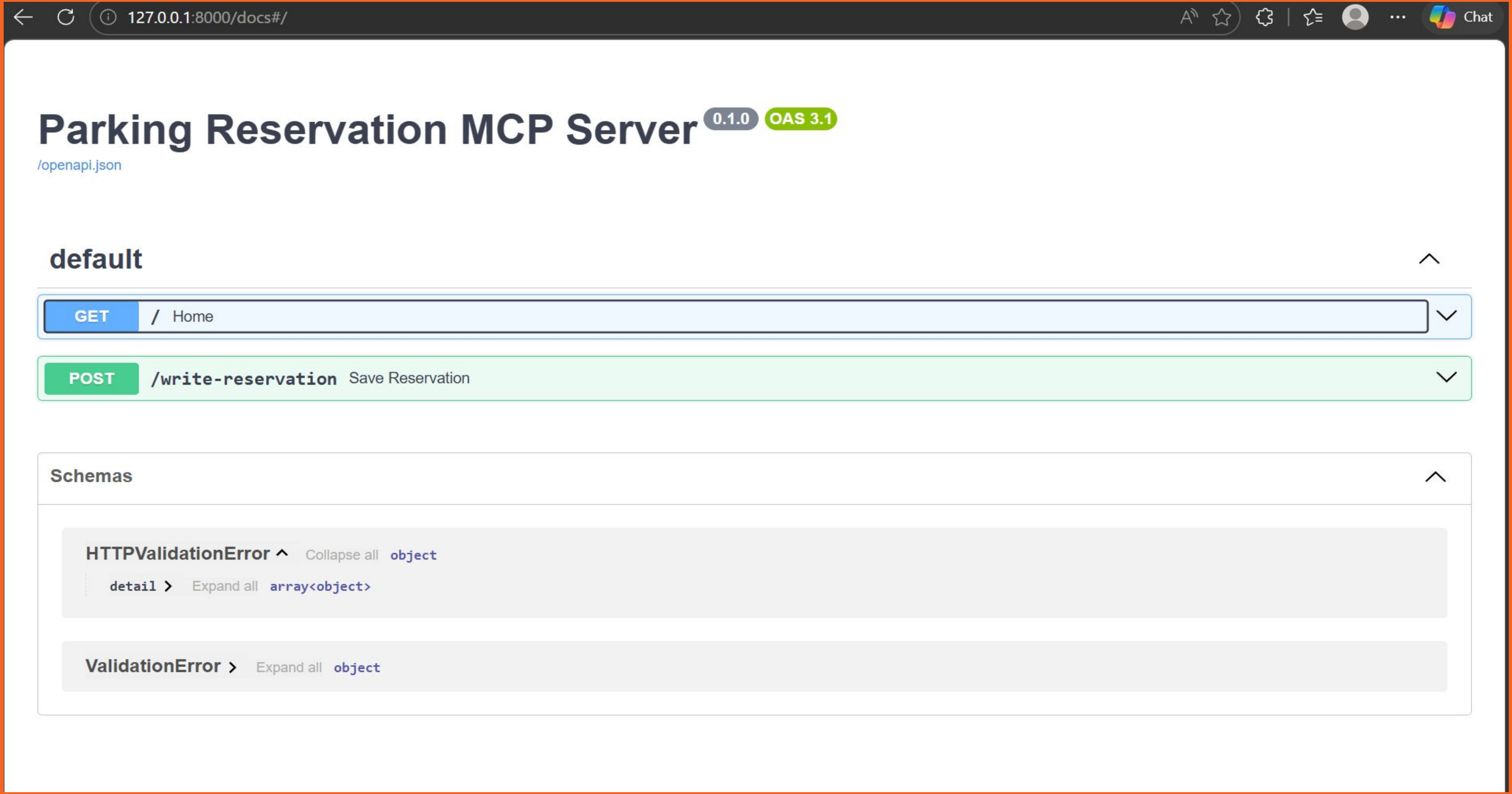
SCREENSHOTS

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER powershell

```
configfile: pyproject.toml
plugins: anyio-4.13.0, langsmith-0.8.5, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 23 items

tests/test_agents.py::test_valid_aadhaar PASSED                                      [  4%]
tests/test_agents.py::test_invalid_aadhaar PASSED                                    [  8%]
tests/test_agents.py::test_valid_license PASSED                                      [ 13%]
tests/test_agents.py::test_invalid_license PASSED                                    [ 17%]
tests/test_agents.py::test_valid_car_number PASSED                                   [ 21%]
tests/test_agents.py::test_invalid_car_number PASSED                                 [ 26%]
tests/test_agents.py::test_valid_date PASSED                                         [ 30%]
tests/test_agents.py::test_invalid_date PASSED                                       [ 34%]
tests/test_agents.py::test_valid_end_date PASSED                                     [ 39%]
tests/test_agents.py::test_invalid_end_date PASSED                                   [ 43%]
tests/test_availability.py::test_standard_availability PASSED                        [ 47%]
tests/test_availability.py::test_vip_availability PASSED                             [ 52%]
tests/test_guardrails.py::test_detect_malicious_prompt PASSED                        [ 56%]
tests/test_guardrails.py::test_safe_prompt PASSED                                    [ 60%]
tests/test_mcp.py::test_write_reservation PASSED                                     [ 65%]
tests/test_mcp.py::test_duplicate_reservation PASSED                                 [ 69%]
tests/test_mcp_api.py::test_health_endpoint PASSED                                   [ 73%]
tests/test_mcp_api.py::test_write_reservation PASSED                                 [ 78%]
tests/test_mcp_api.py::test_missing_field PASSED                                     [ 82%]
tests/test_reservation.py::test_next_missing_field PASSED                            [ 86%]
tests/test_reservation.py::test_no_missing_field PASSED                              [ 91%]
tests/test_retrieval.py::test_recall_at_k PASSED                                     [ 95%]
tests/test_retrieval.py::test_precision_at_k PASSED                                  [100%]

============================================ 23 passed in 0.78s ============================================
```


127.0.0.1:8000/docs#/
Chat
Parking Reservation MCP Server
0.1.0
OAS 3.1
/openapi.json
default
GET
/ Home
POST
/write-reservation
Save Reservation
Schemas
HTTPValidationError
Collapse all
object
detail
Expand all
array<object>
ValidationError
Expand all
object

```
(.venv) PS C:\Users\ShoaibMohdKhan\Desktop\parking-rag-system> pytest tests/test_mcp.py -v
============================================ test session starts =============================================
platform win32 -- Python 3.11.9, pytest-9.0.3, pluggy-1.6.0 -- C:\Users\ShoaibMohdKhan\Desktop\parking-rag-syst
em\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\ShoaibMohdKhan\Desktop\parking-rag-system
configfile: pyproject.toml
plugins: anyio-4.13.0, langsmith-0.8.5, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 2 items

tests/test_mcp.py::test_write_reservation PASSED                                                          [ 50%]
tests/test_mcp.py::test_duplicate_reservation PASSED                                                      [100%]

============================================= 2 passed in 0.11s ==============================================
(.venv) PS C:\Users\ShoaibMohdKhan\Desktop\parking-rag-system>
```

```
============================================= 2 passed in 0.11s =============================================
(.venv) PS C:\Users\ShoaibMohdKhan\Desktop\parking-rag-system> pytest tests/test_mcp_api.py -v
============================================ test session starts ============================================
platform win32 -- Python 3.11.9, pytest-9.0.3, pluggy-1.6.0 -- C:\Users\ShoaibMohdKhan\Desktop\parking-rag-syst
em\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\ShoaibMohdKhan\Desktop\parking-rag-system
configfile: pyproject.toml
plugins: anyio-4.13.0, langsmith-0.8.5, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 3 items

tests/test_mcp_api.py::test_health_endpoint PASSED                                                     [ 33%]
tests/test_mcp_api.py::test_write_reservation PASSED                                                   [ 66%]
tests/test_mcp_api.py::test_missing_field PASSED                                                       [100%]

============================================= 3 passed in 0.21s =============================================
(.venv) PS C:\Users\ShoaibMohdKhan\Desktop\parking-rag-system>
```

THANK YOU

<epam>

India